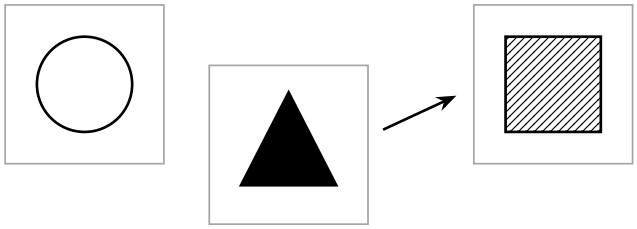
Use the nine tiles, one of each shape and fill. An allowed three uses three different tiles. Its shapes are all the same or all different. Its fills are also all the same or all different. Tile positions do not matter.



allowed three

shapes ○ △ □ all different

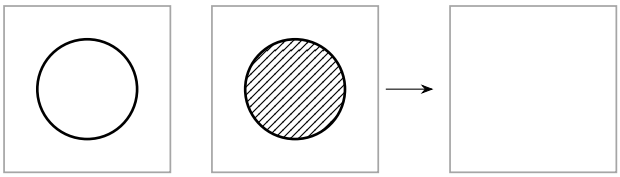
fills □ ■ ▨ all different

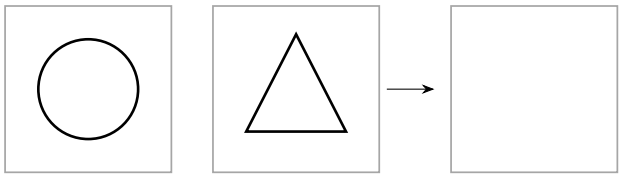
not allowed

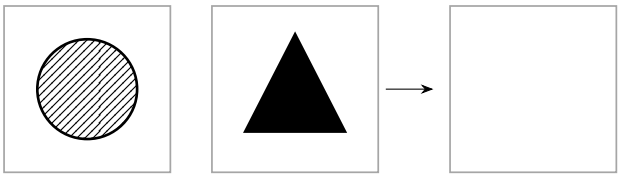
shapes ○ △ □ all different

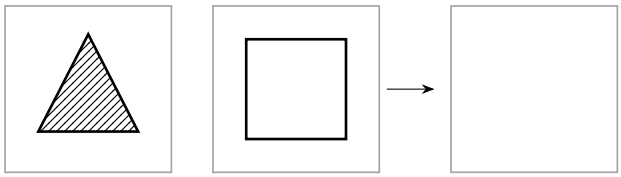
fills □ □ ■ two the same

Problem 1: Complete each pair to make an allowed three. Choose other pairs to give your partner; can any pair have two different answers?



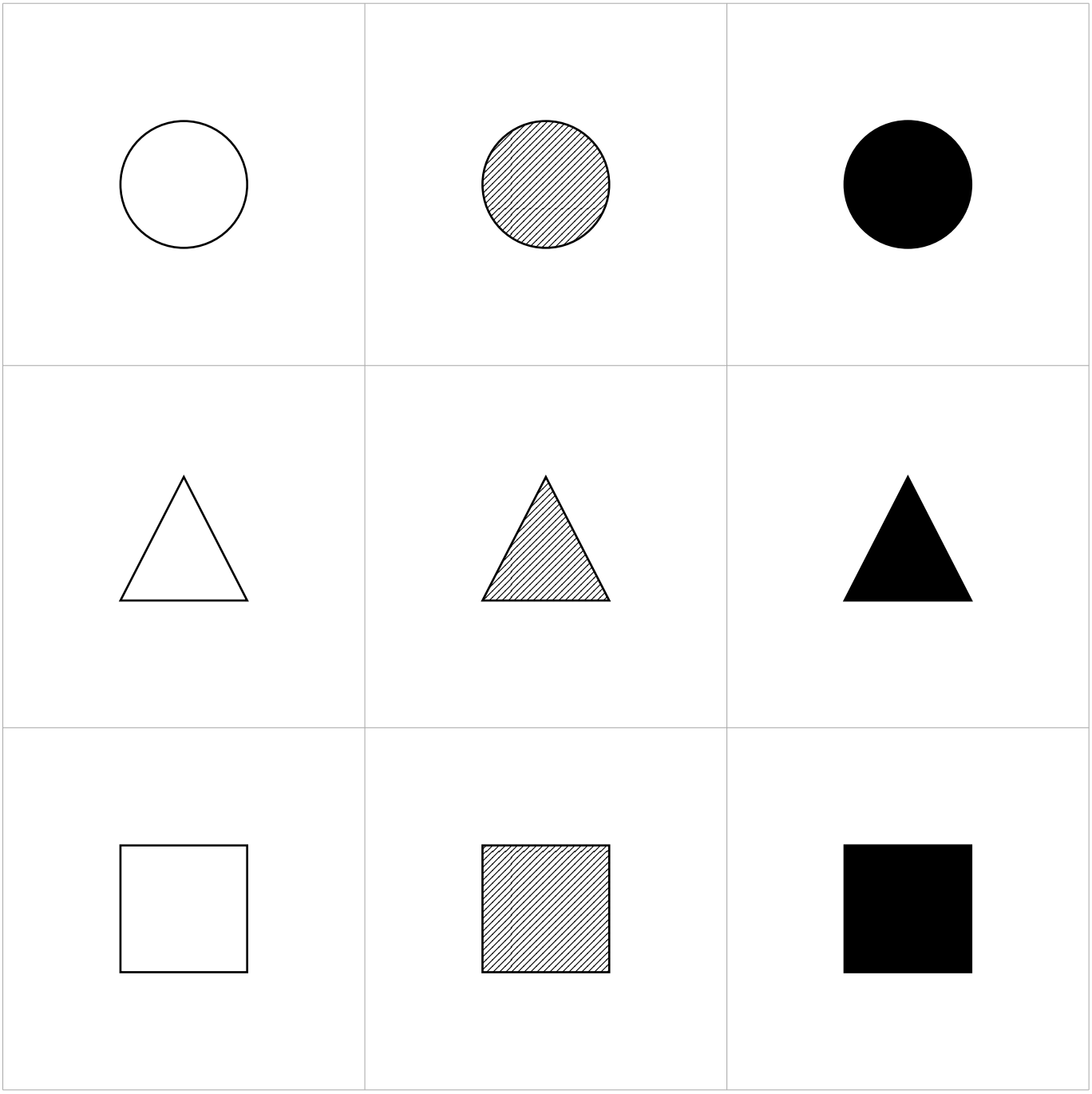




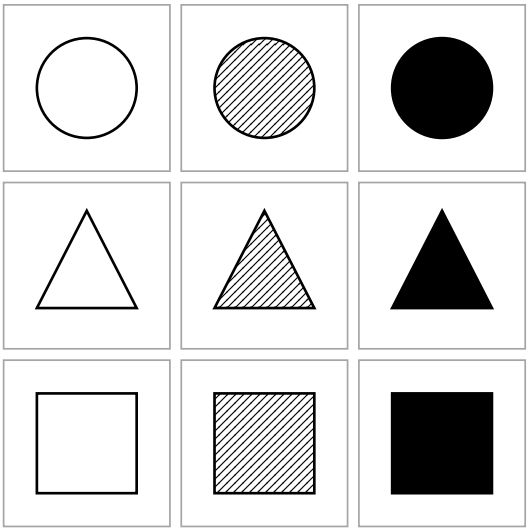


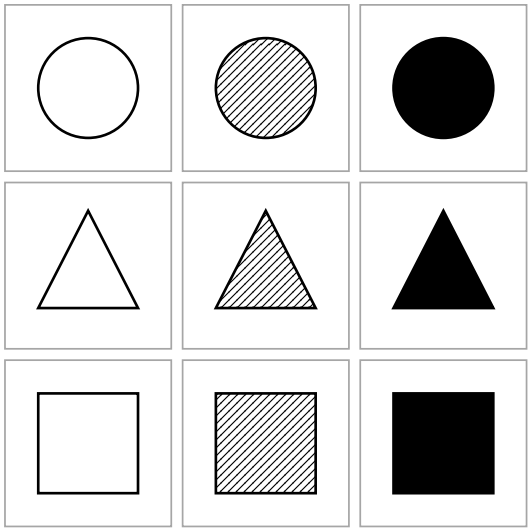
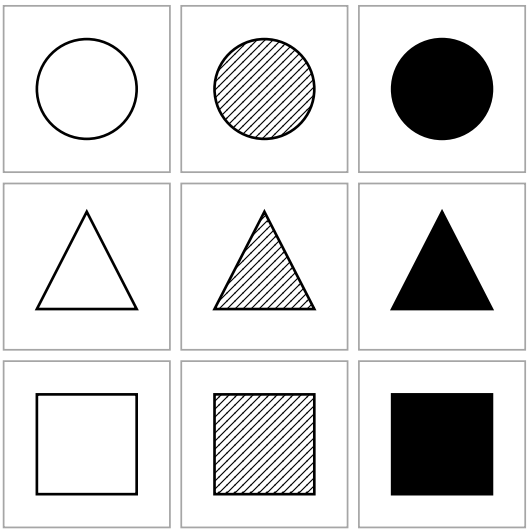
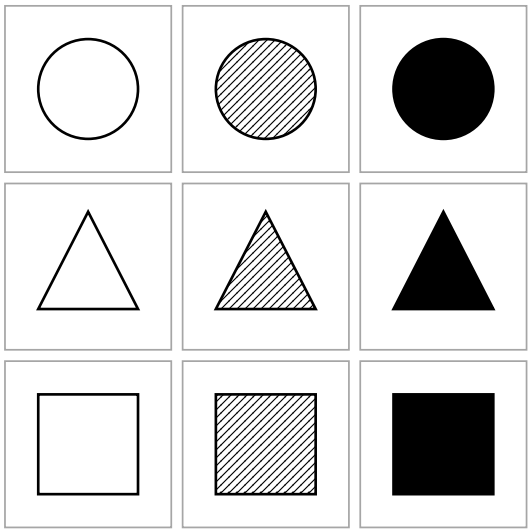
Problem 2: Split all nine tiles into three allowed threes. Find different splits; changing the order of the same three groups does not make a new split.

Problem 3: Choose as many tiles as you can with no allowed three among them. How do you know you cannot keep more?



Problem 4: Choose one tile. Find all the allowed threes that use it, and explain why you have them all.





Problem 5: Start with no tiles chosen. Take turns adding an unused tile to the same collection without making an allowed three. The first player who cannot move loses. Can your choices change how many moves the game lasts?